

Physicochemical Properties

Intrinsic physical and chemical characteristics, such as appearance, boiling point, density, volatility, water solubility, and flammability, are known as physicochemical properties of a substance.

Incorporating information about a compound's physicochemical properties into a QSAR model can enhance the model's ability in predicting the compound's toxicity. This is because these properties influence the compound's interactions with other molecules and its ability to enter and exit cells, which ultimately impacts its toxicity. As a result, these properties are commonly used as descriptors in modeling ecotoxicity, ADME properties, and other toxic properties.

Table 1. QSAR Flex models for physicochemical properties.

Model Name	Version	Model Description	Training Set Size	Descriptors*	Output Units
LogP	3.0	Octanol-water partition coefficient	12645	FRAGS	-
Water Solubility	3.0	Water solubility at 25°C	3800	LogP, MW, FRAGS	gm/L
Vapor Pressure	2.0	Vapor pressure at 25°C	1829	WS, MW, TPSA, E-STATE, FRAGS etc.	Log mm/Hg
Boiling Point	2.0	Boiling points of organic compounds	4890	WS, VP, MW, FRAGS	Degrees Celsius

*LogP – Octanol-water partition coefficient; WS – Water solubility; TPSA – Total polar surface area; TPSA – Total polar surface area, E-STATE – Sum of E-State values on specific atom types; BP – Boiling point; VP – Vapor pressure; MW – Molecular weight; FRAGS – Molecular fragments.

Model Details

LogP: LogP, or the octanol-water partition coefficient, is a metric that measures the distribution of a chemical between nonpolar and polar solvents and reflects its lipophilicity. It is an essential parameter in QSAR models and can provide insight into a chemical's ability to permeate biological membranes, thereby influencing its potential for toxicity. Lipophilicity plays a crucial role in drug absorption, bioavailability, hydrophobic drug-receptor interactions, metabolism, and toxicity, as well as the environmental fate of chemicals. This model has been trained on a dataset of 12645 compounds, using ordinary least square regression as the modeling methodology.

Water Solubility: Water solubility is the maximum amount of a substance that can be dissolved in water at a given temperature. In this case, measured in moles per liter at 25 degrees Celsius. It is important in QSAR and predictive toxicology as it determines the chemical's bioavailability, environmental fate and toxicity, and potential to adsorb to organic matter. It also affects pharmacokinetics of drugs. This model is trained on 3800 compounds using ordinary least square regression as the modeling methodology.

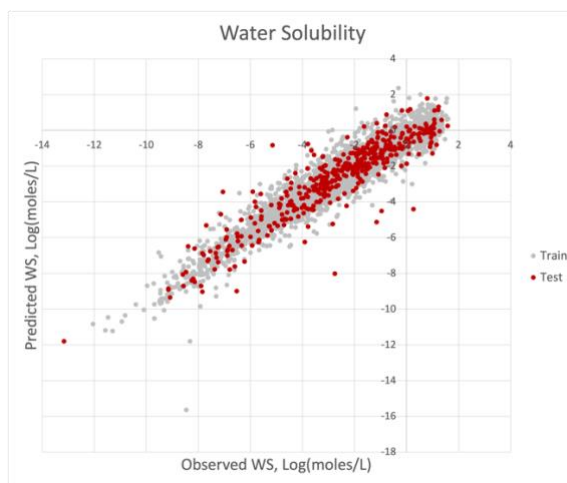
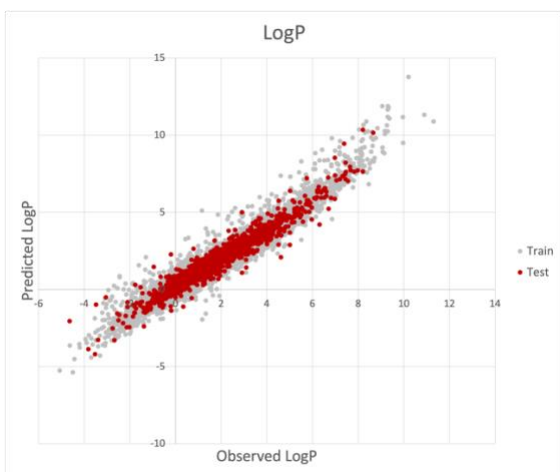
QSAR Flex Model Information

Vapor Pressure: Vapor pressure is the pressure exerted by a liquid's vapor at equilibrium. In this case, we have used vapor pressure data measured at 25 degrees Celsius in mm/Hg. Vapor pressure is an important descriptor in QSAR and predictive toxicology as it indicates a chemical's volatility, affecting its potential for exposure through inhalation, its adsorption to surfaces and its environmental fate in the atmosphere. This model is trained on 1829 compounds using ordinary least square regression as the modeling methodology.

Boiling Point: Boiling point is the temperature at which a liquid turns into a gas at normal atmospheric pressure. It is measured in degree Celsius at 760 mm Hg. It indicates a chemical's physical state and volatility, affecting its potential for exposure through inhalation and its behavior in various conditions. It also helps in determining storage and handling needs. This model is trained on 4890 compounds using ordinary least square regression as the modeling methodology.

Validation of the Models

The models underwent thorough validation through internal cross-validation (CV) and by predicting both the training set and external test sets withheld during the model development phase. The CV process was executed in a 10-fold manner (except the LogP model which was validated 10% out 5 times), where the model was completely rebuilt in every cycle.



QSAR Flex Model Information

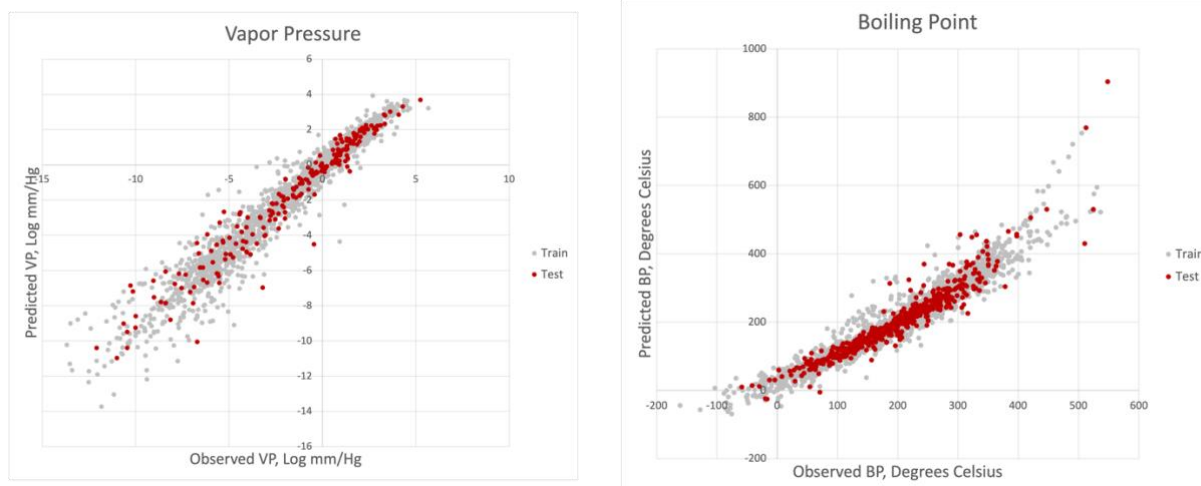


Figure 1. Comparison of experimental and predicted toxicity for the models, for both the training and external test sets.

Table 2: Results of validation of the models.

Model	Train		Test		CV		Train set size	Test set size
	R^2	RMSE	R^2	RMSE	R^2	RMSE		
LogP	0.942	0.441	0.930	0.484	0.931	0.482	12645	1405
Water Solubility	0.907	0.692	0.844	0.965	0.872	0.835	3800	422
Vapor Pressure	0.943	0.860	0.940	0.943	0.901	1.107	1829	203
Boiling Point	0.934	22.365	0.885	33.043	0.924	23.759	4890	543

R^2 – Squared correlation coefficient; $RMSE$ – Root mean squared error; CV – 10% out 10 times cross validation (except the LogP model which was validated 10% out 5 times).

References

1. Chakravarti, S. K., FlexFilters: A Scalable and Flexible QSAR Platform for Addressing Complex and Diverse Types of in Silico Safety Assessment of Chemicals, Poster, Society of Toxicology Annual Meeting, Anaheim, 2020.
2. Mansouri, Kamel, Chris M. Grulke, Richard S. Judson, and Antony J. Williams. "OPERA Models for Predicting Physicochemical Properties and Environmental Fate Endpoints." *Journal of Cheminformatics* 10, no. 1 (December 2018): 10. <https://doi.org/10.1186/s13321-018-0263-1>.